Semantic-DP Distillation: Learning through Semantic Clusters at Small Privacy Budgets

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ABSTRACT

Community-content learning needs shared supervision when labels remain with clients and public proxies are unlabelled. Joint per-example vote reports make whole-client sensitivity grow with proxy size, although related examples can share supervision. Semantic-DP Distillation (SDD) makes public semantic regions the unit of private knowledge exchange: each client casts one class vote per region, and the released regional target supervises every proxy example in that region. For K fixed public regions, the report contains KC coordinates with tight replace-one sensitivity $\sqrt{2K}$. Established distributed integer noise and adaptive accounting provide whole-run client privacy under protected aggregation. On frozen public representations with 10^4 strongly heterogeneous CIFAR-10 clients, a fixed configuration using no public task labels achieves 84.2% accuracy over five seeds at $\varepsilon = 0.1$ and $\delta = 10^{-5}$, versus 42.4% for independent learning and 69.7% for per-example DP-FedCT. The 100-coordinate report gains 14.4 percentage points with 200× fewer coordinates. Separate public-configured controls yield 85.0% for semantic regions versus 48.9% for size-matched random regions. With public configuration selection, a ten-seed Usenet study of 3000 distinct authors reaches 26.3% balanced accuracy at $\varepsilon = 1$, gaining 5.1 points over per-example transfer using 30 rather than 9000 coordinates. Semantic reuse extends the supervision supplied by each bounded private report.

CCS CONCEPTS

• **Security and privacy** → **Privacy-preserving protocols**; • **Computing methodologies** → *Machine learning*.

KEYWORDS

differential privacy, federated learning, distributed DP, knowledge distillation, semantic partition, private aggregation

1 INTRODUCTION

Community-content classifiers can share public examples while users retain labelled post histories and local models on their devices. A coordinator supplies an unlabelled public proxy, a public feature map and a class taxonomy; useful task supervision must come from clients through protected aggregates. Prio and Prochlo provide protected Web sums [5, 9], and browser reporting protects user contributions [3, 11, 13]. For collaborative learning, those sums must carry reusable supervision. The design question is how to preserve that supervision at small whole-client privacy budgets while histories and model parameters remain local.

The gap: related examples can share supervision. Knowledge transfer queries client models on public examples: PATE aggregates teacher votes [21, 22], and federated co-training shares hard labels [2]. Joint per-example voting represents each proxy item separately, with replace-one whole-client sensitivity $\sqrt{2|P|}$, even when task-related examples can share a target. NoiseCluster provides cluster-level label reuse under label privacy [23]. Client collaboration protects an entire local dataset and its adaptively trained model response. The unit of private knowledge exchange therefore determines both report size and sensitivity.

The idea: one released target, many learning examples. Semantic-DP Distillation (SDD) exchanges model opinions over a fixed public semantic partition. Each client votes once per region using pooled predictions (SDD-M) or a public centroid representative (SDD-R); every proxy example inherits its released regional target. A K-region report contains KC coordinates with tight whole-client sensitivity $\sqrt{2K}$. At fixed K, additional proxy examples can enter local learning without enlarging the report or its sensitivity, independently of model parameter dimension. For $K = 10$, $|P| = 2000$ and $C = 10$, 100 report coordinates supply targets to all 2000 examples, compared with 20000 per-example coordinates. Established distributed integer noise [15] and conditional composition protect the complete adaptive release sequence under an aggregate-only interface.

The evidence: client-sourced supervision at 100 coordinates. With frozen public representations and actual 10^4 strongly heterogeneous CIFAR-10 clients, the fixed $K = C = 10$ configuration without public task labels reaches 84.2% **accuracy at** $\varepsilon = 0.1$ **over five seeds**, versus 42.4% for independent local learning and 69.7% for per-example DP-FedCT. A 100-coordinate report gains **14.4 percentage points with** 200× **fewer coordinates**; its paired 95% interval is $[12.2, 16.7]$. The paired local gain is 41.8 points ($[38.0, 45.6]$). Public task labels select neither this configuration nor its training targets.

Separate public-configured controls identify why regional reuse is useful: semantic grouping gives 85.0% accuracy versus 48.9% for size-matched random groups. The **36.1-point gain** ($[32.0, 40.1]$) holds report size, privacy, participation and local computation fixed. Semantic structure increases the usefulness of a fixed-size regional report. On natural Web contributions, a ten-seed, public-configured $K = 5$ study of 3000 distinct Usenet authors gives 26.3% balanced accuracy at $\varepsilon = 1$, versus 21.3% for per-example transfer, with 30 rather than 9000 coordinates. These results connect regional supervision reuse to client-sourced learning and compact release resources.

Contributions.

- **Regional model reports for client-private learning.** A K-region vote interface reuses released targets across public examples. Its KC-coordinate report has tight whole-client sensitivity $\sqrt{2K}$, independent of proxy size and model dimension at fixed K (§3).
- **Client-level privacy for adaptive distillation.** We establish client-level privacy for SDD's complete adaptive learning run under protected aggregation and bounded collusion, applying

established distributed discrete Gaussian accounting to integer regional reports (§4).

- **Semantic reuse as a learning resource.** Client-sourced supervision reaches 84.2% with 100 coordinates without public task labels. Matched structure controls and natural-author results identify how semantic regions support compact knowledge transfer (§5).

2 RELATED WORK

We organise related work by the released object and aggregation interface, which together determine the cost of private knowledge transfer.

Parameter-level release. DP-SGD establishes the clip-and-noise template for example-level privacy [1]; DP-FedAvg extends it to client-level privacy [16, 18] in federated averaging [17]. Secure aggregation and distributed discrete Gaussian noise support this release without a trusted curator [6, 8, 15]. Noise occupies all released model coordinates, making cohort size an important resource. SDD uses the same aggregate-only privacy interface for a compact semantic report. Our clipped-head baseline controls for low-dimensional parameter release using the same public features.

Per-example knowledge transfer. PATE returns noisy arg-max answers from teacher votes on individual queries [21, 22]; federated co-training shares hard labels on a public unlabelled set [2]. Recent distillation work at this venue calibrates and decouples global knowledge for individual clients [25]. Under client-level adjacency, a joint release of per-example vote vectors has sensitivity $\sqrt{2|P|}$. Data-dependent PATE accounting additionally exploits teacher agreement. SDD chooses a public semantic partition before privatisation, replacing the per-example release unit with a designer-selected number of regions. This makes K, rather than $|P|$, the sensitivity variable.

Cluster-level label privacy. NoiseCluster [23] assigns one noisy majority label to an unsupervised cluster and reuses it across its examples. This establishes cluster-level label reuse as a prior design idea. Its adjacency changes one label while holding features fixed, giving sensitivity $\sqrt{2}$ for the cluster-count vector. SDD protects a whole client's features and labels: replacing its trained model can change all K public-region opinions, giving $\sqrt{2K}$. Its contribution is a regional model-report interface for adaptive client collaboration, with distributed aggregation and whole-run client privacy. These different protected units preclude a direct comparison at the same numerical ε.

Aggregate-only infrastructure on the Web. Prio, Prochlo and split-and-mix protocols provide building blocks for private aggregation [5, 9, 14]. Browser-facing systems incorporate budgeted reporting and summary releases [3, 13], while recent Web Conference work uses distributed noise sharing in data markets [27]. Shuffling can also amplify local randomisers under a different accounting route [10, 12]. SDD analyses distributed noise under an aggregate-only interface: anonymity supports that interface, and the accountant operates on the released sum.

Table 1: Release interfaces under replace-one client adjacency, before quantisation and transport sharing. B is the clipping norm.

released object	coordinates	L_2 sensitivity
clipped parameter update	d_{model}	$2B$
joint per-example votes	$\lvert P\rvert C$	$\sqrt{2\lvert P\rvert}$
public-region votes	KC	$\sqrt{2K}$

Sensitivity design and verifiability. Recent Web privacy methods design sensitivity for node-level graph learning [26] and adapt budgets in federated unlearning [24]. SDD bounds an entire client's regional report. Explicit convolution accounting and per-release certificates connect that bound to the implemented aggregate mechanism, applying the established discrete-Gaussian sampler and convolution analysis [8, 15].

3 SEMANTIC-DP DISTILLATION

A public semantic region lets one released target supervise many proxy examples. SDD uses regions as the unit of private knowledge exchange: clients cast one bounded vote per region, and the protected aggregate supplies shared targets for local learning.

Table 1 compares clipping and release resolution. Our compact head baseline shares 128-dimensional features. SDD requires compatible class opinions while keeping each client's parameters local.

3.1 A public partition for knowledge reuse

A fixed public extractor g and public K-means form nonempty regions $P_1, \ldots, P_K$, shared across rounds independently of private data. At fixed K, additional proxy examples increase distillation inputs without increasing reports. Singleton regions at $K = |P|$ recover per-example transfer. Semantic reuse requires compatible task information within a region. Random partitions satisfy the same sensitivity bound; matched experiments test the learning value of their geometry separately.

3.2 One bounded opinion per region

For client model f_i, the regional posterior is

$$\bar{R}_i[k, \cdot] = \frac{1}{|P_k|} \sum_{x \in P_k} f_i(x) \in \Delta^{C-1}. \tag{1}$$

SDD-M votes for the largest component of $\bar{R}_i[k, \cdot]$. SDD-R votes on $f_i(x_k^*)$, where x_k^* is the public point nearest the centroid: public representative queries followed by regional label propagation. Both emit K one-hot rows and propagate targets over identical regions. Fixed-K comparisons distinguish pooled and localized responses while holding sensitivity, report size and partition quality fixed.

Reports lie in a product of K probability simplexes. Direct posteriors also satisfy this bound, with quantisation slack. The evaluated one-hot encoding lies exactly on the integer grid. Common learners isolate release granularity; the interface itself requires compatible class outputs from local models.

3.3 Noisy consensus and local learning

Each client adds independent integer discrete-Gaussian shares to its quantised report. The aggregate-only layer exposes $S = \sum_i (sR_i + G_i)$, and the server projects $S/(sn)$ row-wise onto the simplex to obtain A. For ordinary consensus, the target is

$$\hat{y}(x) = \arg\max_c A[k(x), c]. \tag{2}$$

We also evaluate a fixed class-prior correction: with public region weights $w_k = |P_k|$, subtract $\pi_c = \sum_k w_k A[k,c] / \sum_k w_k$ before taking the arg-max. This uses the released aggregate and public sizes. It can counter a common class preference when clients have narrow local label coverage. The choice between ordinary and corrected consensus is made using public development data, shared across label-release methods, and checked separately in the post-processing ablation. Both preserve privacy by post-processing.

Later-round local batches mix equal private and proxy halves, using hard-label cross-entropy and fixed optimisation steps. Every proxy item inherits its region's target. The confidence-filtered baseline samples its answered subset under separately budgeted screening.

Report size. At $|P| = 2000$, $C = 10$, a $K = 10/20$ release uses 100/200 integer coordinates, versus 20000 per-example coordinates and 1280 head coordinates. Counts measure per-round learning payload before secret sharing and cryptographic overhead.

Choosing resolution without private tuning. The public pilot fixes K, consensus and baseline clipping/screening using public development teachers and simulated deployment noise. The label-free check fixes $K = C$ and centered consensus without public labels. Both estimators remain separate. Final runs aggregate actual client reports and independent shares, calibrated for that same population.

4 THREAT MODEL AND PRIVACY GUARANTEE

4.1 Private units and the aggregation interface

Client i holds its entire local dataset D_i and model f_i. A fixed cohort of n clients participates in each of T rounds. The public proxy P, its partition, feature extractor and release configuration are fixed independently of these private datasets. The adversary observes the noisy sum and every broadcast; local models, individual reports and honest noise shares remain on the client side of the aggregation interface.

A1 (aggregate-only view). Secure additive aggregation reveals the sum of contributions and protocol metadata that is public or independent of protected data, while hiding individual values. Secret sharing or split-and-mix aggregation can instantiate this interface [6, 9, 14]. The privacy accountant operates on the released sum.

A2 (bounded collusion). At most $t < n$ contributors disclose their reports and noise to the server; at least $h = n - t$ independent honest shares remain. The main evaluation uses $t = 0$ and separately tests $t = 0.1n$.

Table 2: Notation. Population and calibration agree in every learning run.

symbol	meaning
n, h	actual participants; honest contributors $h = n - t$
K, C, d	public regions; classes; report dimension
T	release rounds; 3 in the evaluation
s	integer quantisation scale, 2^{16}
b	integer discrete-Gaussian share scale
Δ_q	sensitivity of the quantised sum
ρ_ε	whole-run zCDP allowance

Algorithm 1 One round of Semantic-DP Distillation

client i:
  $\tilde{R}_i \leftarrow$ regional posteriors (mean or representative)
  $R_i[k, \cdot] \leftarrow e_{\arg\max_c \tilde{R}_i[k,c]}$
  $Z_i \leftarrow sR_i$ ▷ exact integer votes
  $G_i[j] \leftarrow \mathcal{N}_{\mathbb{Z}}(0, b^2)$ independently
  submit shares of $Y_i = Z_i + G_i$ modulo 2^{64}
server:
  $S \leftarrow \sum_i Y_i$ ▷ aggregate-only view
  $A \leftarrow \mathrm{Proj}_\Delta(S/(sn))$ row-wise
  extract cluster targets by fixed public post-processing
  broadcast targets for hard-label local distillation

A3 (protected aggregation). Aggregation parties maintain A1 against the server and the stated coalition. The learning experiment implements the sum arithmetic; secure transport is an assumed deployment component.

A4 (compliant contributions). Honest clients use the bounded report and fresh independent noise. Participation is fixed ($q = 1$); a deployment rechecks calibration or aborts if the required honest cohort is unavailable. The guarantee concerns observation and collusion under this interface.

Replace-one client adjacency. Adjacent inputs replace the entire dataset of one honest client. Conditioned on any previous public transcript, that client's report may change arbitrarily within the bounded report space. Other clients use the same transcript and their own local data. Conditional sensitivity and adaptive composition therefore protect the complete server transcript, even though reports evolve over rounds. All evaluated population sizes are actual reporting populations: the averaging denominator and noise calibration use the same n.

4.2 A bounded integer release

Here $\mathcal{N}_{\mathbb{Z}}(0, b^2)$ has mass proportional to $\exp(-z^2/(2b^2))$ on integers. Its scale parameter bounds its standard deviation; its variance is strictly below b^2 [15]. The implementation uses the rejection sampler of Canonne et al. [8], with rational Bernoulli decisions. Modular addition, dequantisation, projection and target extraction are post-processing. Appendix C specifies the sampler and replay protocol.

Lemma 4.1 (Report sensitivity). *For $R_i, R_i' \in (\Delta^{C-1})^K$, $\|R_i - R_i'\|_2 \le \sqrt{2K}$, and this report-space bound is tight. It holds for every fixed public partition, including random partitions.*

Each simplex row has diameter $\sqrt{2}$; opposing one-hot rows attain the $2K$ squared bound. Semantic controls separately test learning quality.

Lemma 4.2 (Quantised sensitivity). *For one-hot reports, $\Delta_q = s\sqrt{2K}$ exactly. For rounded posterior reports in dimension $d = KC$, $\Delta_q \le s\sqrt{2K} + \sqrt{d}$. For an update clipped at norm B, $\Delta_q \le 2sB + \sqrt{d}$.*

Two rounding-error vectors differ by at most one per coordinate, hence by at most $\sqrt{d}$ in norm. For the clipping baseline, define the integer radius $L_B = \lceil sB + \sqrt{d}/2 \rceil$ and enforce this ball after rounding; its certified sensitivity is $2L_B$. The integer guard includes any floating clipping error. The main vote interface incurs no rounding slack. Sensitivity depends on the emitted object, independently of how a client trains its model.

4.3 Explicit convolution and whole-run accounting

Theorem 4.3 (One aggregate release). *Under A1–A4, let $h = n - t$, $b \ge 1$, and*

$$\tau_h(b) = 10 \sum_{j=1}^{h-1} \exp\left(-2\pi^2 b^2 \frac{j}{j+1}\right), \tag{3}$$

$$\rho_r = \frac{\Delta_q^2}{2hb^2} + \frac{d\tau_h(b)}{4}. \tag{4}$$

The aggregate is ρ_r-zCDP for replace-one client adjacency. The empty sum is zero when $h = 1$; otherwise $\tau_h(b) \le 10(h-1)e^{-\pi^2 b^2}$.

After subtracting colluding contributions, h independent discrete-Gaussian shares protect the integer query. Equation (4) is the first branch of Proposition 10 in Kairouz et al. [15], with the actual replace-one sensitivity Δ_q. Additional honest noise and modular reduction preserve this guarantee. This bound accounts for the convolution distribution directly.

Theorem 4.4 (Adaptive whole run). *For T conditionally calibrated releases, $\rho_{\text{total}} \le \sum_{r=1}^{T} \rho_r$. The server transcript is (ε, δ)-DP whenever*

$$\rho_{\text{total}} + 2\sqrt{\rho_{\text{total}} \log(1/\delta)} \le \varepsilon. \tag{5}$$

The composition includes all broadcasts [7]. Public feature extraction, public-only configuration selection and distillation of released labels consume no additional client budget. Offline benchmark accuracy and clean-report diagnostics measure the mechanism; they are not additional outputs claimed to satisfy its privacy guarantee.

Calibration that can be checked numerically. Writing $L = \log(1/\delta)$, set $\rho_\varepsilon = (\sqrt{L+\varepsilon} - \sqrt{L})^2$ and choose

$$b = \left\lfloor \frac{\Delta_q}{\sqrt{2h\rho_\varepsilon/T}} \right\rfloor + 1. \tag{6}$$

We then verify the full corrected bound, including $d\tau_h/4$, against ρ_ε/T. The upward integer step provides slack; every experimental release has $\tau_h < 10^{-30}$. Machine-readable certificates record $n, h, b, \Delta_q, d, \log \tau_h$ and the recomposed privacy bound. For $K = 10$, $T = 3$, $\varepsilon = 0.1$, $\delta = 10^{-5}$, the continuous reference scale on the sum is 372.50: residual scales are approximately 3.725, 0.3725 and 0.03725 at actual $n = 100$, 1000 and 10000. These are different participating populations in the learning experiment.

Participation and resolution. Since $\mathrm{Var}(G_i[j]) < b^2$, the full-cohort mean has per-coordinate standard deviation bounded by $b\sqrt{n}/(sn)$. With $t = 0$, this is close to

$$\frac{\sigma}{n}, \qquad \sigma = \sqrt{TK/\rho_\varepsilon}. \tag{7}$$

At a fixed accountant, per-example and cluster votes therefore have noise-scale ratio $\sqrt{|P|/K}$ up to integer calibration. A larger cohort supplies more reports as well as a larger averaging denominator. Its evolving consensus is measured at each actual population size in the scaling experiment.

Collusion tolerance. Replacing n with $h = n - t$ in calibration retains the corrected guarantee after the coalition subtracts its shares. With all n shares present, the noise-scale upper bound increases by approximately $\sqrt{n/(n-t)}$, or 1.054 for 10% collusion tolerance. Integer rounding and τ_h remain in each certificate. This quantifies the extra noise required by the stated coalition model; the paired utility experiment measures its learning effect.

5 EVALUATION

The central comparison measures the task supervision clients supply through an unlabelled proxy. Matched region controls identify the role of semantic geometry, and accuracy–report-size operating points characterize release resources. Natural-author results connect these choices to the allocation of supervision.

5.1 Experimental setup

Public features and private learners. CIFAR-10 and MNIST use a frozen public DINOv2 ViT-S/14 extractor [20], followed by a 128-component PCA fitted on the public proxy alone and per-example feature normalisation. Each client's separately initialised softmax classifier trains on its own 600 draws with replacement from the private image pool, with a Dirichlet label mixture $\alpha \in \{0.1, 1.0\}$. There are 45000/53000 eligible private CIFAR-10/MNIST images after excluding the 5000 public development images and the proxy. Overlap between constructed clients is possible; the protected unit is one complete constructed dataset.

The image proxy contains 2000 unlabelled examples. Public validation and final test use different halves of the official test split after removing proxy examples: 3967/4033 CIFAR-10 examples and 5000/5000 MNIST examples. Public development labels train the pilot teachers. Client-report methods use private labels and released proxy targets; public-supervised references are defined below. The selector omits final-test arrays; final evaluation follows configuration and schedule freezes.

Natural-author Web content. The 20 Newsgroups corpus supplies Usenet posts and From-header author identifiers. We define six public content families: comp, rec, sci, talk, alt/soc and misc. Each client comprises all training posts of one selected author identifier; no author is duplicated to enlarge the cohort. There are 4249 eligible private authors, with a median of one training post. Separate public authors supply 1500 proxy and 1500 development

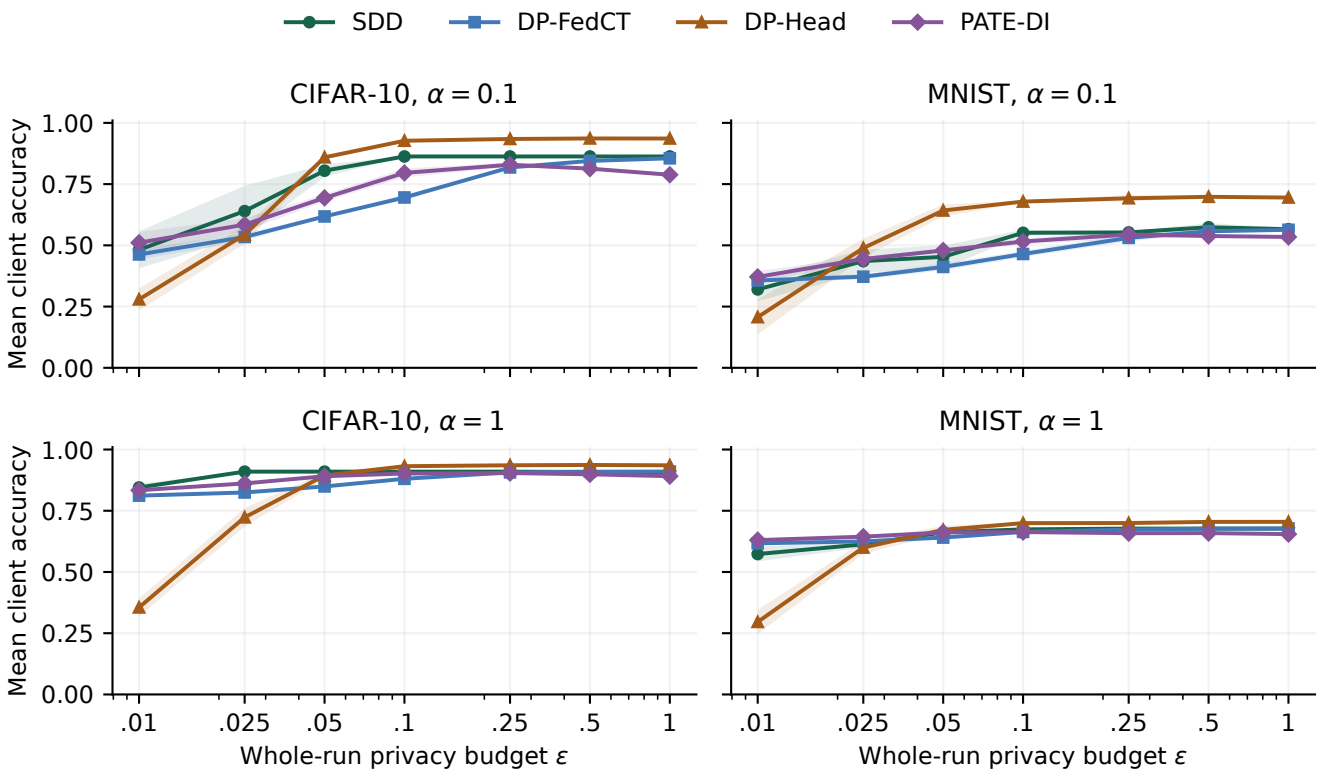


Figure 1: Public-configured privacy–utility curves, actual 10^4 reporting clients, three seeds with sample SD. SDD denotes SDD-M. Methods share public features and private optimisation; PATE-DI includes protected screening costs. Five-seed $\varepsilon = 0.1$ estimates appear in Table 4.

posts. Validation and test contain 1747 and 1737 posts from disjoint held-out authors absent from both private and public training authors. Headers and quoted lines are removed from model input. TF-IDF is fitted on public proxy/development text, and its 128-dimensional SVD is fitted on the proxy. The predeclared primary Web metric is mean client balanced accuracy over the six families; standard accuracy is also retained in the artifact.

Identical local resources. All private learners use three rounds of 60 SGD steps, batch size 128, learning rate 0.1, momentum 0.9 and weight decay 5×10^{-4}. Round 1 is private; later batches mix 64 private and 64 labelled proxy items. PATE samples answered queries, using private-only batches when none qualify. Client weights, momentum and draws are independent. Final metrics average third-round client models; DP-Head additionally evaluates its final broadcast and the local endpoint preceding it. All three releases are charged, with no extra adaptation or cohort-dependent training steps.

Privacy and configurations. Image curves use actual $n = 10^4$ at seven budgets $\varepsilon \in \{0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1\}$, $\delta = 10^{-5}$. Additional cohorts are 100/1000 image clients and 100/1000/3000 distinct authors. Every release samples one noise share per actual contributor, certified under §4. A public pilot selects $K \in \{5, 10, 20, 50, 100\}$ and ordinary/centered consensus by validation transfer, using public teachers and five deployment-noise repeats. Configurations freeze before private test metrics; each actual cohort forms its own consensus.

Matched release interfaces. DP-FedCT retains FedCT's hard-label co-training [2] but replaces its native local randomised response with the same aggregate-only distributed-DP layer as SDD. It reports all $|P|C$ one-hot coordinates, uses the same local trainer, and receives the same public consensus correction. This matches the privacy unit while varying granularity.

DP-Head privately averages clipped $128 \times C$ classifier updates, adds exact noise shares, and broadcasts the resulting head. Its public pilot selects the clipping radius from six public-teacher norm quantiles. Rounded updates are guarded by integer radius $L_B = \lceil sB + \sqrt{128C}/2 \rceil$ and use sensitivity $2L_B$. It shares the feature map and private optimisation budget.

PATE-DI uses two protected stages: noisy maximum vote confidence screens queries, then noisy vote counts answer a capped subset. Its public pilot tunes three thresholds, two answer-cap fractions and two screening-budget fractions. Screening and answers are both included in data-independent Rényi accounting [19, 22].

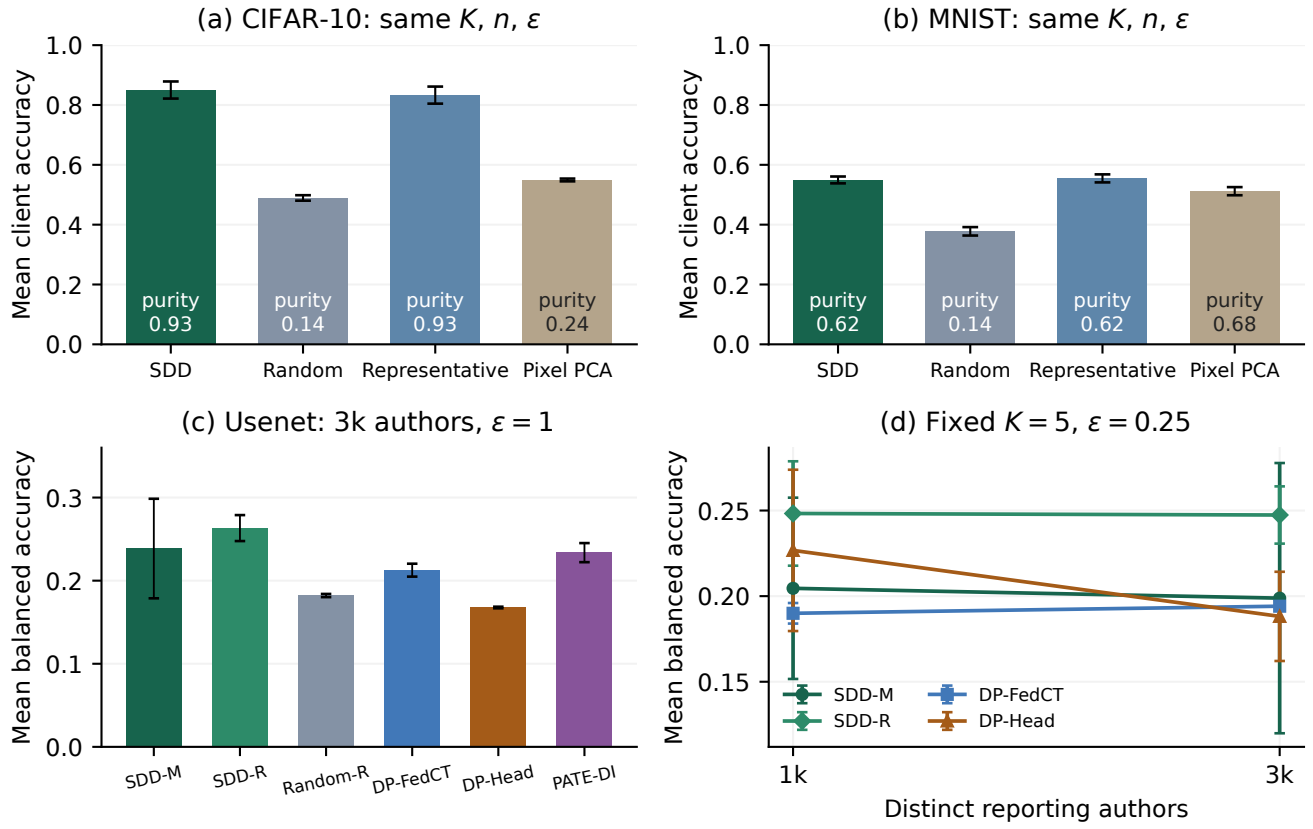


Figure 2: Public-configured semantic controls and natural-author transfer, means with sample SD. (a,b) Images: five seeds, actual $n = 10^4$, $\alpha = 0.1$, $\varepsilon = 0.1$, fixed K and learner features; purity is public. (c) Usenet: 3k authors, $\varepsilon = 1$. (d) Fixed $K = 5$, $\varepsilon = 0.25$ separates participation from resolution. Web uses ten fresh seeds. M/R share sensitivity and report size.

Protected maximum computation requires a richer secure functionality than SDD's additive sum; the evaluation grants this ideal functionality to the baseline. The configured protocol and tighter conversion are in Appendix D.

Replications and uncertainty. Complete budget curves use independent population/training/noise seeds 801–803. The predeclared $\varepsilon = 0.1$ image and Web comparisons and semantic controls use five seeds, 801–805. Curves show means and sample standard deviations. Independent confirmations use ten fresh seeds (8101–8110) and frozen configurations: actual $n = 1000$ CIFAR-10 at 0.1, and Web cohorts 1000/3000 at fixed $K = 5$ and budgets 0.25/1. Both estimators retain frozen settings. Paired 95% intervals use seeds as replications. For paired differences d_j, intervals are $\bar{d} \pm t_{0.975,m-1} s_d/\sqrt{m}$, with $m = 5$ or 10 seeds; gains are computed before rounding and measured in percentage points. The independent ten-seed group supplies the primary estimate at overlapping settings; three-seed curves retain the initial scan. Table 12 retains the initial, confirmation and pooled groups. The artifact contains every result, configuration and schedule.

Knowledge-source references. Local-only uses the same clients and 180 steps, with private data throughout. Public-only trains a head on all existing public development labels, using the same feature map. Public validation chooses learning rate from 0.03, 0.1, 0.3 and training length from 60, 180, 600 steps over five public seeds, before test access. We report both the 180-step reference and the selected 600-step model. Local+Public holds the client's 180-step schedule and proxy mixing fixed but obtains proxy labels from that public model, isolating the source of shared supervision. These references transmit no private model or report. Public-configured runs make the same labelled development pool available to the public pilot and public-supervised references. The no-public-task-label condition fixes $K = C$ and centered consensus; public task labels select neither settings nor training targets. Images use $\alpha = 0.1$, $\varepsilon = 0.1$. Public-configured Web confirmation uses $K = 5$ selected with labelled public teachers; the fixed check uses $K = 6$. Both estimators share each rule, with no routing. Benchmark labels serve offline evaluation. The two conditions jointly vary resolution and selection resources.

Table 3: Client-sourced supervision without public task labels, actual 10^4 clients, $\alpha = 0.1$, $\varepsilon = 0.1$, $\delta = 10^{-5}$, five seeds. Fixed $K = C = 10$ and centered consensus; 180 client steps. Accuracy is percent mean ± sample SD. SDD uses 100 report coordinates per client per round, DP-FedCT 20000, Local-only zero. Gain and paired 95% interval are percentage points relative to DP-FedCT.

task	Local-only	DP-FedCT	SDD-M	gain	paired 95% interval
CIFAR-10	42.4 ± 0.1	69.7 ± 1.6	**84.2 ± 3.0**	14.4	[12.2,16.7]
MNIST	33.3 ± 0.1	46.2 ± 1.1	**51.6 ± 1.8**	5.3	[4.1,6.6]

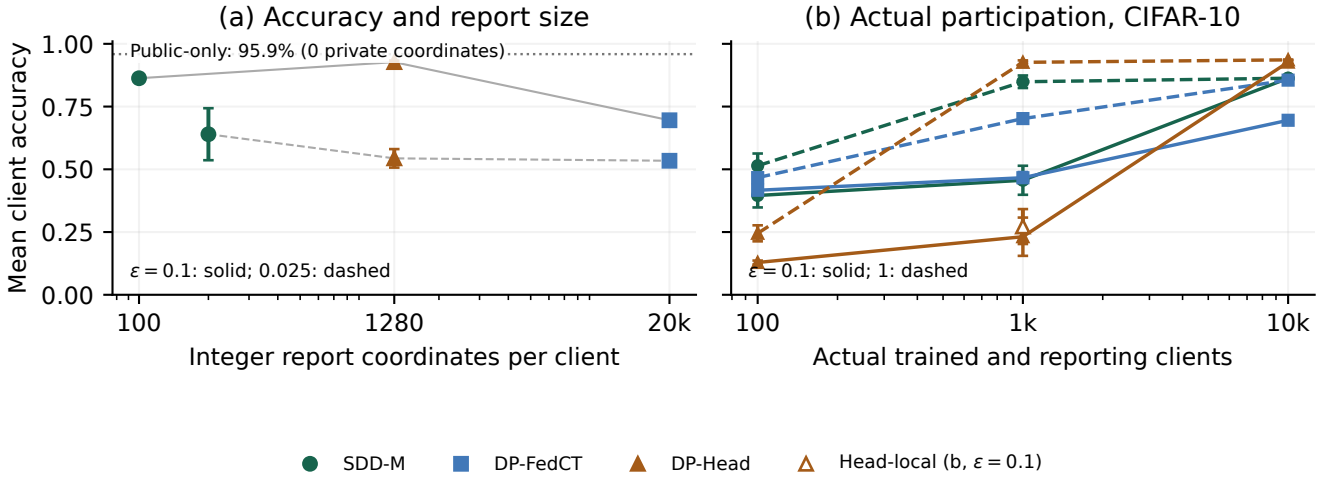


Figure 3: Public-configured CIFAR-10 release resources, $\alpha = 0.1$, means with sample SD. (a) Actual 10k clients, three seeds; dotted line: public-only head with all 5000 development labels, 600 steps and zero private coordinates. (b) Actual cohorts, fixed client steps; $\varepsilon = 0.1$ at 1k uses ten fresh seeds, other points three. Open triangle is DP-Head's matched local endpoint. Solid/dashed curves use $\varepsilon = 0.1/0.025$ in (a) and $\varepsilon = 0.1/1$ in (b); public settings are fixed per cohort.

5.2 Client-sourced supervision without public task labels

Regional client reports raise strongly heterogeneous CIFAR-10 accuracy from 42.4% independent local learning to 84.2% at $\varepsilon = 0.1$, using 100 report coordinates. With actual 10k clients, fixed $K = C = 10$ and centered consensus, five seeds give 84.2% ± 3.0%, versus 69.7% ± 1.6% for per-example DP-FedCT and 42.4% ± 0.1% for Local-only (Table 3). The paired gains are 14.4 points (95% interval [12.2, 16.7]) over per-example transfer and 41.8 points ([38.0, 45.6]) over independent learning. A released regional target supervises every proxy item in its region: 100 coordinates supply targets to all 2000 proxy examples with 200× fewer coordinates than the per-example interface.

The same fixed rule gives MNIST 51.6%, versus 33.3% local and 46.2% per-example accuracy. Public task labels determine neither the configuration nor the training targets; private client reports supply task supervision through the unlabelled proxy. The complete fixed-rule image and Web conditions appear in Table 9.

5.3 Semantic structure and release resolution

At matched report size, privacy budget and local computation, semantic grouping improves CIFAR-10 accuracy by **36.1 percentage points** over size-matched random regions. This separate public-configured, five-seed comparison at $10k/\varepsilon = 0.1$ gives 85.0% versus 48.9%, with paired interval [32.0, 40.1] (Figure 2(a)). Their public purities are 0.925 and 0.139. Pixel-PCA regions give 54.9% at purity 0.240, using the same learner features. These controls show that semantic geometry increases the usefulness of a fixed-size regional report; every partition satisfies the same $\sqrt{2K}$ sensitivity bound.

MNIST similarly gives 55.0% for semantic means versus 37.8% for random groups. Centroid representatives reach 83.3% on CIFAR-10 and 55.5% on MNIST through the same compact interface (Figure 2(a,b)). Regional reuse supports both pooled and representative model opinions. Public geometry is task-specific: pixel-PCA regions can have higher MNIST purity, while the public visual representation supplies the stronger CIFAR-10 partition. Table 7 relates these geometries to first-release pseudo-label correctness, noise flips and downstream learning.

Regional transfer also forms useful low-budget operating points. In the public-configured, three-seed strongly heterogeneous CIFAR-10 scan, the first *tested* budget reaching 80% is $\varepsilon = 0.05$, compared with $\varepsilon = 0.25$ for per-example DP-FedCT: a 5× difference between measured operating budgets. Figure 1 shows all seven budgets and both heterogeneity levels; Table 10 gives the exact CIFAR grid. Five-seed $\varepsilon = 0.1$ comparisons across the four image settings are in Table 4.

Table 4: Predeclared image $\varepsilon = 0.1$ comparisons over five seeds. Accuracy is a percentage; gain and paired 95% interval are percentage points relative to DP-FedCT.

task (α)	SDD	gain	interval
CIFAR (0.1)	85.0	15.3	[12.4,18.2]
CIFAR (1)	91.0	2.9	[2.8,3.0]
MNIST (0.1)	55.0	8.7	[8.3,9.1]
MNIST (1)	67.6	1.2	[0.8,1.6]

5.4 Participation and release resources

Regional reports and clipped-head averaging offer different accuracy–report-size operating points. At $10k/\varepsilon = 0.1$, the public-configured regional interface reaches 85.0% with 100 coordinates; the matched local head reaches 91.5% with 1280, and its global broadcast reaches 92.7%. Both local endpoints use 180 steps (Table 5). The regional report provides a 12.8× smaller learning payload while retaining separately trained client models.

At $1k/\varepsilon = 0.1$, the independent ten-seed public-configured estimate is 45.6% with 200 coordinates, versus 27.2% for the local head and 23.1% for its global broadcast. The paired local-head gain is **18.4 points** ($[12.1, 24.8]$), with 6.4× fewer coordinates. Per-example DP-FedCT gives 46.7% with 20000 coordinates, placing regional transfer at a 100× smaller report size. Figure 3 places accuracy alongside report coordinates and actual participation, with fixed client steps and each cohort's own consensus. Its public-configured 1k rule uses $K = 20$; the fixed no-task-label rule instead uses $K = 10$, yielding 45.5% with 100 coordinates versus 46.6% with 20000 per-example coordinates (Table 9).

5.5 Natural-author transfer and supervision allocation

With public-configured resolution $K = 5$, regional transfer reaches 26.3% **balanced accuracy** for 3000 distinct authors at $\varepsilon = 1$, gaining **5.1 points** over per-example transfer with 30 rather than 9000 coordinates. Ten seeds give DP-FedCT 21.3%, with paired interval $[3.6, 6.5]$. Local-only gives 17.0%; the paired local gain is 9.3 points ($[8.2, 10.4]$). Matched random representatives give 18.2%, so semantic assignment adds 8.1 points ($[7.0, 9.2]$) at the same query count, budget, propagation rule and estimator (Figure 2(c), Table 6).

Regional resolution allocates shared supervision. At the same $K = 5$, SDD-M gives 23.9%. With five regions for six families, each release represents at most $\min(K, C) = 5$ class targets. In releases 1/2 that supervise subsequent learning, SDD-R allocates 98.7% of the 1500 proxy targets to comp/rec/sci; SDD-M allocates 46.1% to talk/alt–soc/misc. Both give every proxy item a target, with distinct class-recall profiles: R's comp/rec/sci recalls are 78.4%, 44.2% and 28.4%, and its other-three-family mean is 2.3%, versus M's 22.0%. Per-run minimum recalls are 1.8% for R and 2.3% for M: each seed first averages evaluated client models' recalls, then takes the six-class minimum, before the ten-seed summary (Table 11). Both estimators' macro gains coexist with uneven per-run class recall. Local models also learn from private examples and targets that can change across releases.

DP-Head gives 16.8% globally and 18.3% at the matched local endpoint; R's paired gains are 9.6 points ($[8.4, 10.7]$) and 8.1 points ($[6.9, 9.2]$). DP-Head's comp target share evolves from 64.7% to 95.4% to 99.8% across broadcasts. The public pilot selects balanced validation accuracy, while client cross-entropy is unweighted; these are the observed supervision allocations under the frozen clipping setting.

Participation at fixed resolution. At fixed $K = 5$ and $\varepsilon = 0.25$, Figure 2(d) compares 1000/3000 distinct authors with ten seeds per point, separating participation from resolution in the natural-author task.

Taxonomy resolution without public task labels. At $3k/\varepsilon = 1$, the fixed $K = C = 6$ rule gives M 20.3%, R 17.6%, DP-FedCT 21.3% and Local-only 17.0%. M adds 3.3 local points ($[0.8, 5.7]$) (Table 9). This taxonomy-based rule and the public-selected $K = 5$ condition jointly vary resolution and selection resources, so their difference does not isolate a public-label effect. The image results establish the core no-task-label supervision gain; the natural-author results characterize the allocation produced by each specified regional rule.

5.6 Supervision resources and whole-run privacy

Public task labels provide a direct route to supervision. The public-configured pilot and public-model references access the same labelled development pool. Training on its 5000 image labels gives CIFAR-10 95.6% at 180 steps and 95.9% at 600; MNIST gives 84.9/89.0%. With 1500 public Web labels, balanced accuracies are 65.6/68.2%. Public-source proxy labels give local CIFAR-10 87.0% and Web 41.0%. These models use zero private report coordinates (Table 5). The fixed no-task-label experiment measures supervision supplied by client reports when the labelled-public training route is unavailable.

Whole-run client privacy follows §4's aggregation assumptions. At $\varepsilon = 1$, 10% collusion calibration increases the mean-noise bound by about 1.054; both CIFAR-10 settings retain their five-seed accuracy. Appendices B–C and the artifact link integer calibration and implementation checks to frozen configurations and outputs.

6 CONCLUSION

SDD makes public semantic regions the unit of private knowledge exchange. One released regional target supervises many proxy examples; fixed K controls the KC-coordinate report and whole-client sensitivity $\sqrt{2K}$. With frozen public representations, actual 10k strongly heterogeneous CIFAR-10 clients and $\varepsilon = 0.1$, the fixed configuration without public task labels reaches 84.2% using 100 coordinates, gaining 14.4 points over per-example transfer with 200× fewer report coordinates. Separate matched structure controls identify the value of semantic geometry, and natural-author results characterize regional supervision allocations. Regional target reuse extends client-sourced supervision through a compact, protected knowledge-transfer interface.

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A SUPPLEMENTARY RESULTS

The exact CIFAR-10 budget grid complements Figure 1. All MNIST/Web grids and per-seed records are included in the anonymous artifact.

Consensus correction. Ordinary consensus gives the same five-seed CIFAR-10 results as the selected mode at $\varepsilon = 0.1$. MNIST's correction changes mean accuracy by 0.4 and 0.5 points in the two regimes. The structure controls and shared post-processing comparison address distinct design choices.

B PROOFS AND INTEGER CALIBRATION

Proof of Lemma 4.1. Squared Euclidean distance decomposes over rows: $\|R - R'\|_2^2 = \sum_{k=1}^{K} \|R[k,\cdot] - R'[k,\cdot]\|_2^2$. For simplex vectors p, p', $\|p\|_2^2 \le 1$, $\|p'\|_2^2 \le 1$ and $\langle p, p' \rangle \ge 0$, so $\|p - p'\|_2^2 \le 2$. Distinct one-hot rows attain equality. Repeating them across all K rows attains $2K$. Statistical independence between rows is unnecessary. □

Proof of Lemma 4.2. Write $\mathrm{round}(sR) = sR + e$, with $|e_j| \le 1/2$. Then $\|e - e'\|_2 \le \sqrt{d}$, giving $\Delta_q \le s\sqrt{2K} + \sqrt{d}$. One-hot votes have zero rounding error. Two updates clipped to a radius-B ball differ by at most $2B$, giving $2sB + \sqrt{d}$ after rounding. The implemented guard confines integer reports to radius $L_B = \lceil sB + \sqrt{d}/2 \rceil$, hence sensitivity $2L_B$. It checks squared integer norms exactly in the recorded numeric range and replaces an out-of-ball report with zero. Each conditional update is clipped relative to the same preceding public broadcast. □

Proof of Theorems 4.3 and 4.4. Condition on a public transcript and a coalition's disclosed contributions. The remaining

Table 5: Public-configured supervision-source and parameter-endpoint references, percent mean ± sample SD (Web: balanced accuracy). Images use $\alpha = 0.1$, $\varepsilon = 0.1$, five seeds at 10k and ten elsewhere; Web uses ten. M/R denote image/Web SDD. All share public features. Pub heads use all 5000 image/1500 Web development labels, learning rate 0.3 and zero private report coordinates; Local+Pub uses their proxy targets. Client methods use 180 steps. Head-local precedes the third aggregation, retaining two broadcasts with all three releases charged.

	supervision sources					parameter endpoints	
task / variant, cohort	SDD	Local	Local+Pub	Pub (180)	Pub (600)	Head-global	Head-local
CIFAR (M), 10k	85.0 ± 2.9	42.4 ± 0.1	87.0 ± 0.0	95.6 ± 0.1	95.9 ± 0.1	92.7 ± 0.5	91.5 ± 0.4
MNIST (M), 10k	55.0 ± 1.1	33.3 ± 0.1	64.1 ± 0.1	84.9 ± 0.1	89.0 ± 0.1	67.3 ± 1.0	65.0 ± 1.0
CIFAR (M), 1k	45.6 ± 5.8	42.3 ± 0.4	87.0 ± 0.1	95.6 ± 0.1	95.9 ± 0.1	23.1 ± 7.6	27.2 ± 7.0
Usenet (R), 1k, 0.25	24.8 ± 3.0	17.1 ± 0.0	41.0 ± 0.4	65.6 ± 0.4	68.2 ± 0.5	22.7 ± 4.7	22.8 ± 2.6
Usenet (R), 3k, 1	26.3 ± 1.6	17.0 ± 0.0	41.0 ± 0.3	65.6 ± 0.4	68.2 ± 0.5	16.8 ± 0.1	18.3 ± 0.3

Table 6: Fixed-resolution Usenet confirmation, $K = 5$, ten fresh seeds. SDD-M pools regional posteriors; SDD-R queries the public centroid representative; Random-R uses matched random groups and representatives. Balanced accuracy ± sample SD.

n	ε	SDD-M	SDD-R	Random-R	DP-FedCT	DP-Head	PATE-DI
1000	0.25	0.205 ± 0.053	0.248 ± 0.030	0.185 ± 0.004	0.190 ± 0.006	0.227 ± 0.047	0.195 ± 0.010
1000	1	0.172 ± 0.067	0.254 ± 0.018	0.181 ± 0.001	0.196 ± 0.002	0.199 ± 0.036	0.203 ± 0.006
3000	0.25	0.199 ± 0.079	0.247 ± 0.017	0.182 ± 0.001	0.194 ± 0.002	0.188 ± 0.026	0.204 ± 0.011
3000	1	0.239 ± 0.060	0.263 ± 0.016	0.182 ± 0.002	0.213 ± 0.008	0.168 ± 0.001	0.234 ± 0.011

Table 7: Semantic attribution at actual $n = 10^4$, $\alpha = 0.1$, $\varepsilon = 0.1$, five seeds. Purity is public; pseudo-label accuracy and flips use the first release. Full controls at both budgets and heterogeneity levels are in the artifact.

task	grouping	K	purity	pseudo	flips	accuracy
CIFAR	SDD	10	0.925	0.906	0.020	0.850 ± 0.029
CIFAR	random	10	0.139	0.121	0.660	0.489 ± 0.009
CIFAR	representative	10	0.925	0.876	0.120	0.833 ± 0.029
CIFAR	weak	10	0.240	0.222	0.380	0.549 ± 0.005
MNIST	SDD	20	0.624	0.552	0.210	0.550 ± 0.011
MNIST	random	20	0.145	0.116	0.680	0.378 ± 0.014
MNIST	representative	20	0.624	0.556	0.200	0.555 ± 0.013
MNIST	weak	20	0.683	0.513	0.340	0.512 ± 0.014

Table 8: Original ResNet-18 diagnostic, actual/calibrated $n = 100$, fixed seed42, matched trainer and exact integer noise.

ε	interface	accuracy
0.1	SDD	0.223
0.1	DP-FedCT	0.255
1	SDD	0.389
1	DP-FedCT	0.255

integer query has sensitivity Δ_q and retains at least h independent scale-b noise shares per coordinate. Proposition 10 of Kairouz et al. [15] gives the zCDP upper bound in Eq. (4). Its first branch suffices for all reported certificates. Since $j/(j+1) \geq 1/2$, $\tau_h \leq 10(h-1)e^{-\pi^2 b^2}$. Modulo reduction and later target construction are post-processing. The same conditional bound holds at every round, so adaptive composition sums the bounds. Equation (5) converts the sum to client-level (ε, δ)-DP. □

Strict numerical slack. The implementation increases the integer scale beyond the Gaussian reference and verifies $\Delta_q^2/(2hb^2) + d\tau_h/4 \leq \rho_\varepsilon/T$. It stores $\log[10(h-1)e^{-\pi^2 b^2}]$ and conservatively uses 10^{-30} as an upper bound when the exponential is smaller. Thus underflow never silently discards the correction. The experiment artifacts certify each stage and its composed budget. For 10% collusion tolerance, both h and its correction are recomputed. Ninety-digit recomputation verifies all composed certificate bounds; the largest retained convolution upper bound is 10^{-30}.

Ring arithmetic. Reports and shares are added modulo 2^{64}; the signed representative is dequantised for learning. Privacy is preserved under this modular map even when wraparound occurs. Utility wraparound is controlled using the sub-Gaussian tail of the discrete Gaussian [15]: for a scalar noise sum G, $\Pr(|G| \geq a) \leq 2e^{-a^2/(2nb^2)}$. A union bound across released coordinates and rounds, with signal bounded by the quantised report range, certifies the recorded configurations. Its log probability is below -3.5×10^{18} across primary releases.

Table 9: No-public-task-label check; image rows use $\alpha = 0.1$, $\varepsilon = 0.1$. Taxonomy resolution $K = C$ and centered consensus are fixed for every task without using public task labels for training or configuration; benchmark labels are used for offline evaluation. The feature maps and all 180 client steps are unchanged. Mean ± sample SD in percent; five seeds for 10k image cohorts and ten fresh seeds elsewhere. Both Web estimators are shown without routing.

task, cohort	K	SDD-M	SDD-R	DP-FedCT	Local
CIFAR, 10k	10	84.2 ± 3.0	–	69.7 ± 1.6	42.4 ± 0.1
MNIST, 10k	10	51.6 ± 1.8	–	46.2 ± 1.1	33.3 ± 0.1
CIFAR, 1k	10	45.5 ± 6.6	–	46.6 ± 2.3	42.3 ± 0.4
Usenet, 1k, 0.25	6	18.8 ± 4.0	15.0 ± 2.7	19.0 ± 0.6	17.1 ± 0.0
Usenet, 3k, 1	6	20.3 ± 3.4	17.6 ± 1.6	21.3 ± 0.8	17.0 ± 0.0

Table 10: CIFAR-10: actual $n = 10^4$, three independent seeds. Mean accuracy ± sample SD; all public-selected budgets.

ε	K	SDD	DP-FedCT	DP-Head	PATE-DI
CIFAR-10, $\alpha = 0.1$					
0.01	5	0.482 ± 0.076	0.463 ± 0.021	0.279 ± 0.045	0.511 ± 0.044
0.025	20	0.640 ± 0.104	0.534 ± 0.016	0.544 ± 0.037	0.584 ± 0.026
0.05	10	0.805 ± 0.026	0.618 ± 0.002	0.859 ± 0.008	0.694 ± 0.015
0.1	10	0.863 ± 0.000	0.695 ± 0.012	0.927 ± 0.006	0.796 ± 0.018
0.25	10	0.863 ± 0.000	0.818 ± 0.003	0.934 ± 0.003	0.829 ± 0.009
0.5	10	0.863 ± 0.000	0.845 ± 0.005	0.936 ± 0.001	0.813 ± 0.006
1	10	0.863 ± 0.000	0.855 ± 0.003	0.936 ± 0.001	0.788 ± 0.009
CIFAR-10, $\alpha = 1$					
0.01	20	0.845 ± 0.016	0.812 ± 0.002	0.355 ± 0.037	0.834 ± 0.006
0.025	10	0.910 ± 0.000	0.825 ± 0.002	0.723 ± 0.038	0.862 ± 0.001
0.05	10	0.910 ± 0.000	0.849 ± 0.002	0.894 ± 0.014	0.891 ± 0.001
0.1	10	0.910 ± 0.000	0.880 ± 0.000	0.932 ± 0.003	0.902 ± 0.002
0.25	10	0.910 ± 0.000	0.907 ± 0.001	0.936 ± 0.002	0.904 ± 0.001
0.5	100	0.908 ± 0.000	0.910 ± 0.000	0.937 ± 0.001	0.899 ± 0.001
1	100	0.908 ± 0.000	0.910 ± 0.000	0.936 ± 0.001	0.891 ± 0.000

Table 11: Usenet family recall and protected target allocation, 3k/$\varepsilon = 1$, ten seeds, public-selected configuration ($K = 5$ for SDD). Target % and N_c apply to SDD-M/R, DP-FedCT and DP-Head, which label all 1500 proxy items. Target % divides talk/alt–soc/misc target counts by 1500; N_c counts distinct target classes. Both average releases 1/2 within each seed, then ten seeds. Recalls first average evaluated models within each seed; min recall takes the six-class minimum before summarizing seeds (percent mean ± SD). Other recall columns are seed means.

method	comp	rec	sci	talk	alt/soc	misc	target %	N_c	min recall
Local	35.5	20.3	20.2	11.4	6.8	8.1	–	–	6.8 ± 0.3
SDD-M	41.9	14.6	20.7	27.2	21.2	17.6	46.1	3.8	2.3 ± 0.8
SDD-R	78.4	44.2	28.4	2.1	2.2	2.6	1.3	3.0	1.8 ± 0.4
DP-FedCT	57.3	24.7	24.3	9.2	5.5	6.6	41.4	6.0	5.5 ± 0.3
PATE-DI	64.3	28.0	21.7	14.0	5.8	6.5	–	–	5.7 ± 0.4
Head-global	100.0	0.6	0.0	0.0	0.0	0.0	1.7	4.6	0.0 ± 0.0
Head-local	99.0	3.7	1.5	1.5	2.0	2.0	–	–	1.4 ± 0.3
Local+Pub	78.7	58.1	36.3	41.4	25.1	6.6	–	–	6.6 ± 0.2
Public-only	85.1	81.0	56.6	67.8	62.7	56.1	–	–	55.7 ± 1.2

C REPRODUCIBILITY AND PUBLIC SELECTION

The anonymous supplementary archive has a root `README.md` and `reproduce.py`. Run `python reproduce.py verify` to check inputs, frozen configurations, all numerical tables and stored outputs; `python reproduce.py smoke` replays representative fixed cases. `EXPERIMENT_INDEX.csv` maps each condition to its seeds, configuration, role and result. Feature bundles, candidate scores, raw logs, exact integer certificates and complete grid tables are included. Public selection omits test arrays; configuration hashes are checked against the frozen manifests.

The frozen image extractor is DINOv2 ViT-S/14 from the official public checkpoint [20]; its hash is recorded. All images use 224×224 RGB bicubic resizing and the checkpoint's normalisation.

Table 12: SDD-M seed reconciliation. Within each task, initial and fresh groups have identical public K, consensus mode, feature map, local schedule and budget. The pooled estimate includes every run while retaining group identity. Corresponding conclusions use the independent ten-seed group.

setting	seed group	accuracy (%)
CIFAR, 1k	initial 3	56.0 ± 9.8
CIFAR, 1k	fresh 10	45.6 ± 5.8
CIFAR, 1k	pooled 13	48.0 ± 7.8
Usenet, 1k	initial 3	27.6 ± 3.4
Usenet, 1k	fresh 10	20.5 ± 5.3
Usenet, 1k	pooled 13	22.1 ± 5.7

PCA and feature normalisation are fixed public preprocessing; application to private examples is point-wise. Public K-means uses five initialisations, at most 100 iterations, seed 1729, and deterministic nonempty-cluster repair. Random grouping preserves the semantic partition's exact region sizes. The feature control changes the partition to raw-pixel PCA while retaining the same client learner and feature map. The representative control queries the proxy point closest to each public centroid.

Public pilot teachers use seed 1729, disjoint development data, and the same round-1 learner as the final experiment. Candidate K and correction modes use mean-posterior opinions and are scored by nearest-public-centroid label transfer on validation data, averaged over noise seeds 91001–91005. Exact ties prefer smaller K, then ordinary consensus. Six DP-Head clipping candidates are public norm quantiles $0.5, 0.75, 0.9, 0.95, 0.99, 1.0$; they use five noise repeats. PATE-DI settings use two fixed noise repeats per public candidate. These public design surrogates neither use private client data nor consume their privacy budget. Per-budget choices are retained alongside every candidate score, enabling independent verification of the selection rule. The representative comparisons retain those same K/consensus choices; no private result selects between M and R. The no-public-label check instead uses fixed taxonomy resolution and centered consensus. Proxy labels enter its offline diagnostics, while training targets come from the protected client sum.

Exact noise and experimental replay. The sampler uses a discrete-Laplace rejection proposal and rational Bernoulli $\exp(-\gamma)$ decisions as in Canonne et al. [8]. Rejection from ideal independent random bits and arbitrary-precision fallbacks retain the exact integer distribution and its tail. The executable uses ChaCha20 pseudorandom bit streams under the corresponding cryptographic random-source assumption, with fresh secret keys drawn from the operating system for deployment. The privacy proof uses independent private random coins. Recorded benchmark seeds reproduce offline utility calculations and diagnostics, rather than defining a publicly seeded deployment privacy transcript. Sampler checks compare zero-event probabilities, moments and tails with analytic integer-Gaussian probabilities; the learner is checked against scalar SGD and one-client noninterference before final tests.

Resource and protocol scope. Experiments use one RTX 5090 GPU and 16 CPU noise-sampling threads. Vectorisation trains every client independently and samples every client's share; cached round-1 states accelerate paired comparisons. Private models are not transmitted. The utility simulator evaluates the noisy sum directly and assumes the protected transport of A1–A4. Reported clean pseudo-label metrics and test accuracies are offline benchmark measurements, outside the claimed deployment transcript.

D BASELINE ACCOUNTING AND TRAINING

For DP-FedCT, the public singleton partition is exactly the per-example vote interface. SDD and this variant share initial local states, private batches, local optimiser, budget conversion and post-processing. For DP-Head, each client clips its update relative to the preceding public global head and enforces the public integer-ball radius after rounding; the server averages the protected integer updates, broadcasts the new head and generates proxy targets from this public model. All clients adopt that head and reset momentum. Each round still has 60 local steps. We evaluate both the final broadcast and the clients immediately before that broadcast, retaining the first two protected heads, with no extra local steps. The third release remains charged for both endpoints.

For PATE-DI, let $v(x)$ be the aggregate class-count vector. Protected screening forms noisy $\max_c v_c(x)$ for all $|P|$ queries. Its replace-one L_2 sensitivity is $\sqrt{|P|}$. Thresholds are $0.1, 0.2, 0.3$ in mean-vote units; answer caps are $0.25|P|$ or $0.5|P|$; screening receives 0.25 or 0.5 of the per-round privacy allowance. The remaining allowance answers at most M queries with joint sensitivity $\sqrt{2M}$. Thresholding and selecting the highest noisy confidences are post-processing of screening. The answer sensitivity holds conditionally on every selected subset. Both stages use the corrected distributed integer-noise bound and compose over T rounds. The tighter data-independent Rényi conversion [4] is

$$\varepsilon \le \inf_{a>1} \left[a\rho_{\text{total}} + \frac{\log(1/(a\delta))}{a-1} + \log(1-1/a) \right]. \tag{8}$$

This conversion is granted to PATE-DI in addition to protected screening. The evaluated view includes noisy confidences and selected noisy count vectors. Data-dependent PATE accounting requires additional protected analysis of private consensus and accounting outputs; it is a distinct protocol scope.

Original-architecture corroboration. The full ResNet-18 diagnostic uses actual/calibrated $n = 100$, CIFAR-10 $\alpha = 0.1$ and 6000 replacement draws per client. Two continuation rounds each use 20 epochs; the three protected releases use $\varepsilon = 0.1, 1$. It fixes $K = 10$ and uses a singleton partition for DP-FedCT. Both use the same local GPU trainer, exact noise, and individually trained cached seed-42 round-1 states. Evaluation uses the proxy-excluded first half of test. At $\varepsilon = 0.1$, regional/per-example accuracy is 22.3/25.5%; at 1, it is 38.9/25.5%. These architecture-specific operating points are separate from the main frozen-feature population study.